



ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG QUỐC TẾ
VNU-INTERNATIONAL SCHOOL

ECO CHARGE

GREEN STEPS – CLEAN ENERGY 

ECO CHARGE

BƯỚC CHÂN XANH – NĂNG LƯỢNG SẠCH 



Source: Smart City Spotlight / VTV24

<https://youtu.be/RFA9PvRqwwk?si=UaED1uiUXVKrcxbO>



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I START UP
2025

ECO CHARGE

BƯỚC CHÂN XANH – NĂNG LƯỢNG SẠCH

*Smart charging mat
harvest energy from footsteps*

ECO CHARGE

GREEN STEPS – CLEAN ENERGY



CHARGING
STATION



CHARGE

MENTOR ECOCHARGE

TEACHER PHAN THAI TRUNG

- Former Lecturer, Faculty of Information Technology, National University of Civil Engineering.
- Advisory Board Member, Northern Agriculture Startup Club; Executive Board Member, Vietnam IoT Startup Network.



CEO

Ngô Minh
Châu



COO

Doãn Thị
Phương Thảo



CMO

Nguyễn Thị
Phương Thảo



CFO

Tô Ngọc Minh



CTO

Trần Quang
Tiệp



Problem +
Solution

Description

Business
model

Market

ESG/SDG

Finance

Lack of outlets



Power outages



Aesthetic loss



Charging point overload



Problem +
Solution

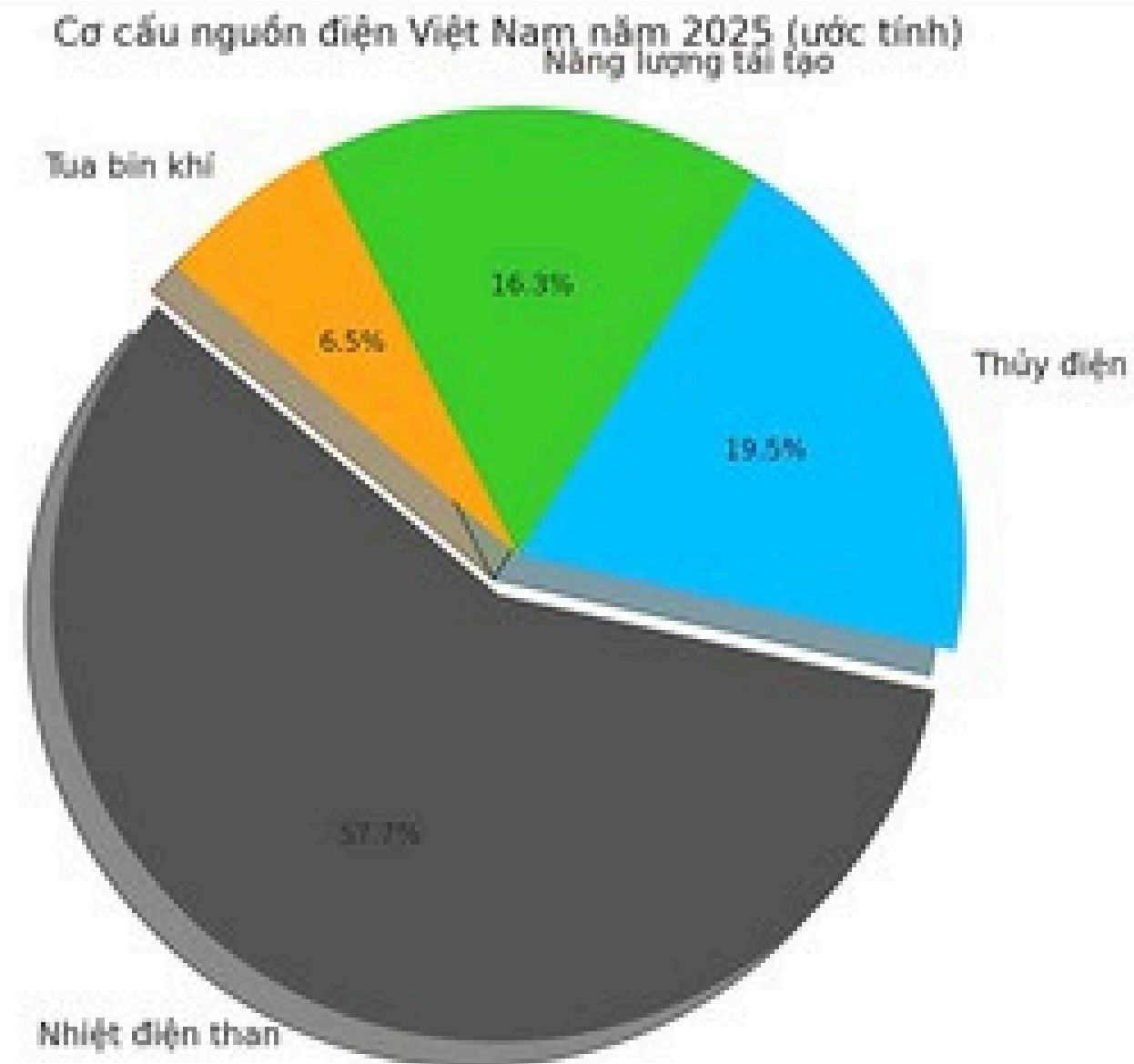
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Business
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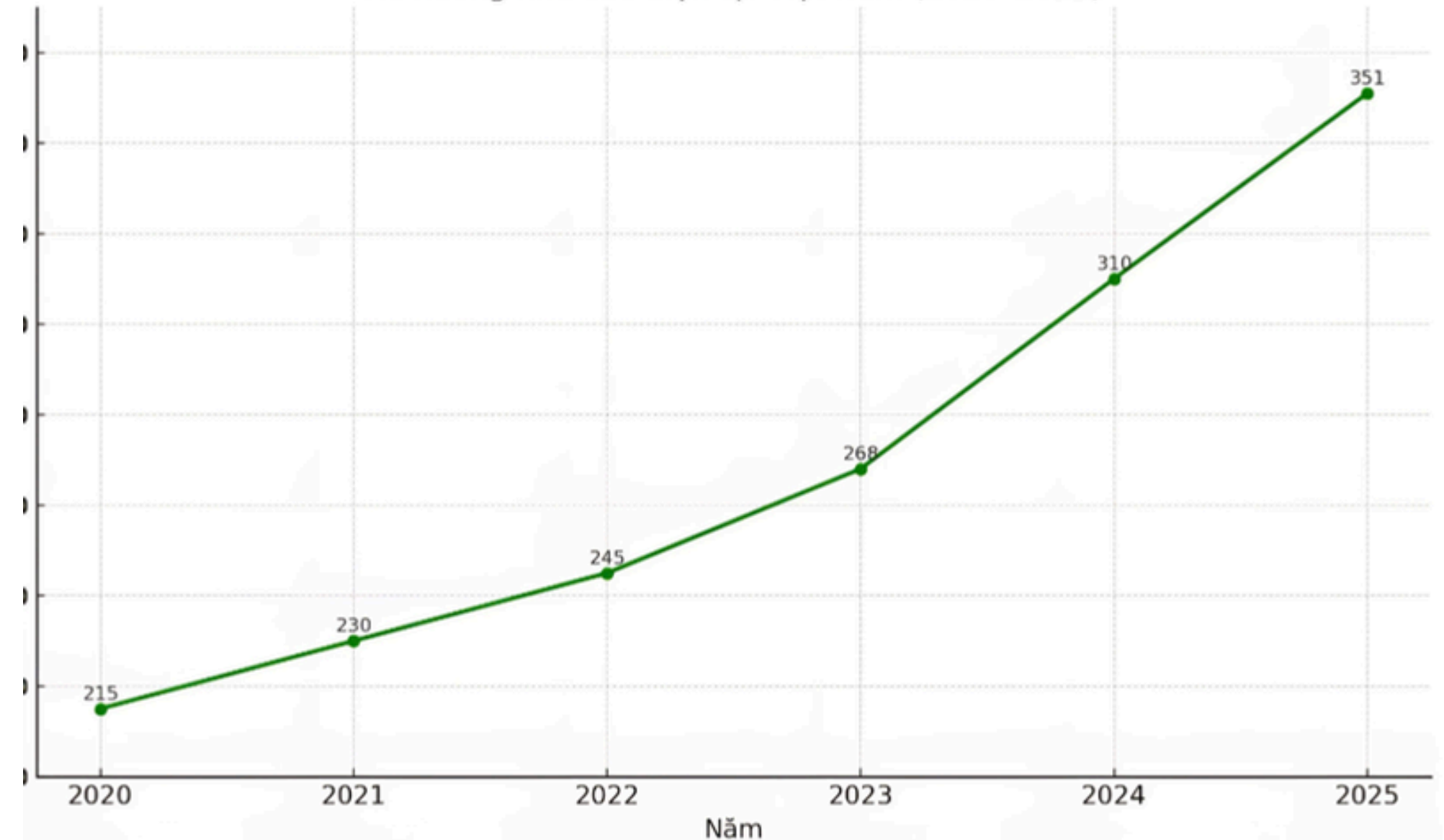
ESG/SDG

Finance



- Vietnam's power generation mix is dominated by coal-fired thermal power.

Xu hướng nhu cầu điện tại Việt Nam (2020-2025)



- Global power demand is projected to grow at an average rate of 2.2% annually (IEA, 2023).

Problem + Solution

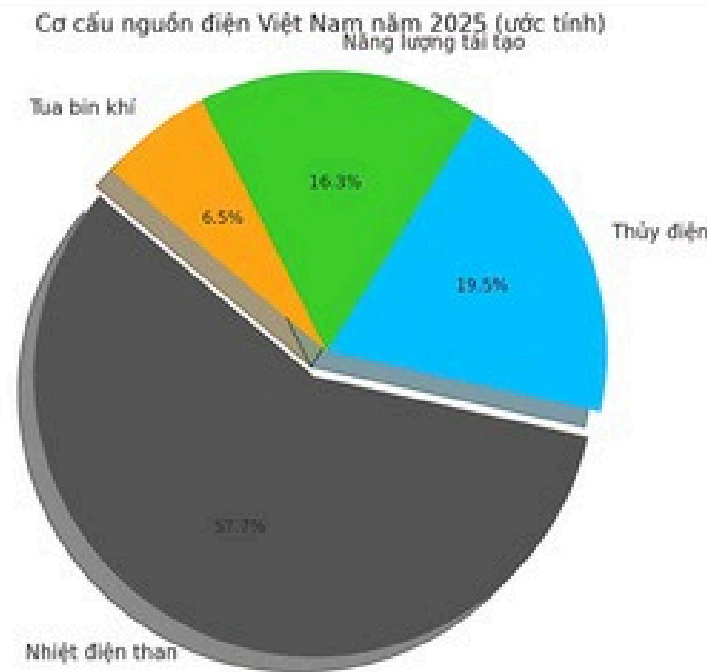
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Business model

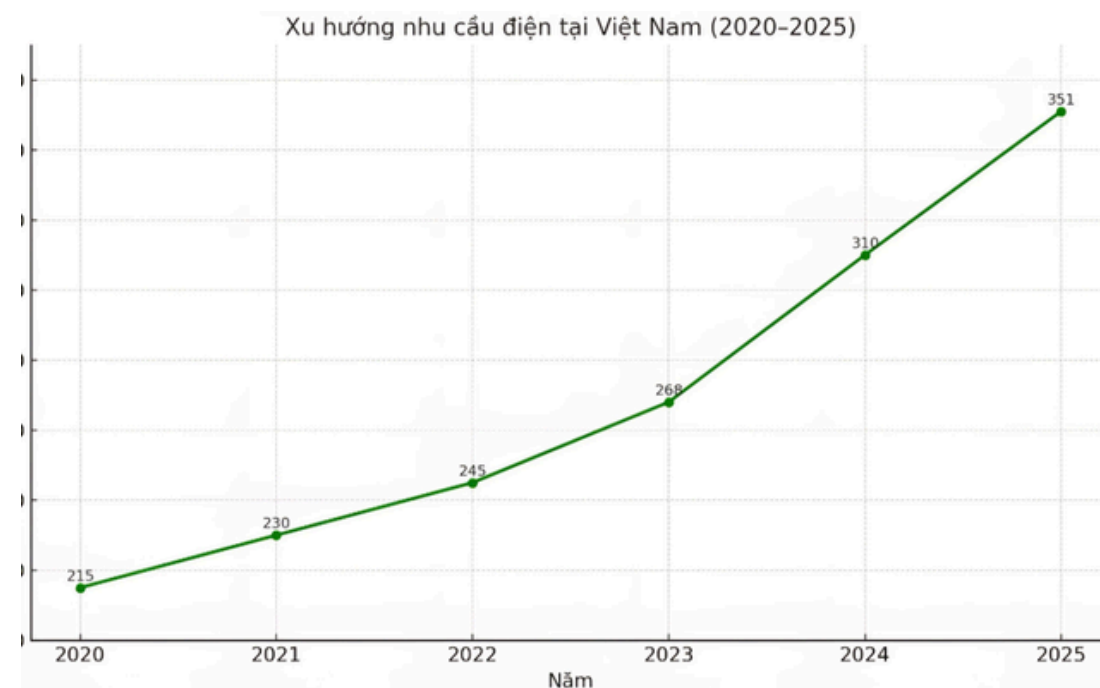
Market

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- Vietnam's power generation mix is dominated by coal-fired thermal power.



=> Prioritizing electric vehicles and smart city development.

==> A surge in electricity demand, while oil and gas reserves are depleting.

- Global power demand is projected to grow at an average rate of 2.2% annually (IEA, 2023).

Problem +
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GREEN – SMART – SUSTAINABLE

Convenient
Charging

Power
Saving

Piezoelectric
technology



CHARGING
STATION



CHARGE

Problem +
Solution

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VALUE PROPOSITION

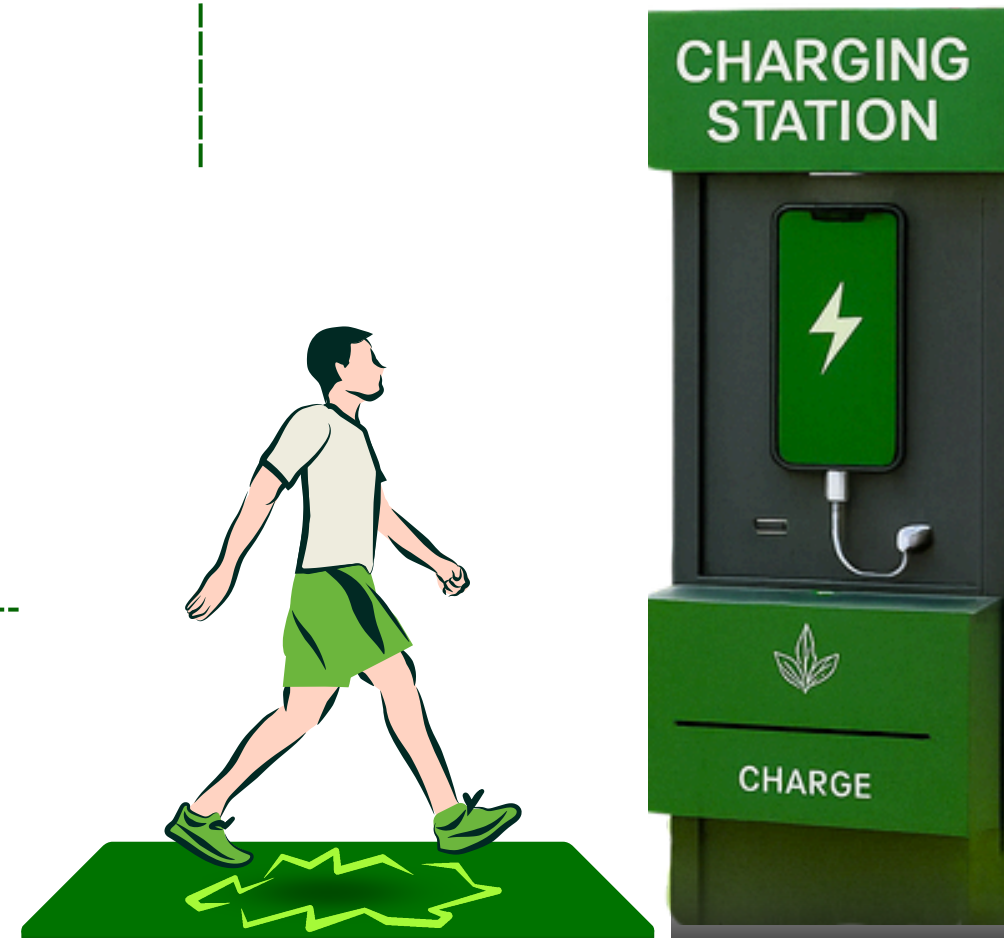
Free Charging &
Off-grid
Autonomy

"Power-it-
yourself"
Experience

Green Branding &
Communication

Easy installation,
scalable modular
design

Footfall Analytics



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WHAT?

EcoCharge is a **smart kinetic charging mat** that:
+ Captures mechanical energy from **every step** taken
+ Transforms into **green electricity** for free public phone charging



WHEN?

When users need to charge their devices, but:

- No power outlets are nearby
- Forgot to carry a power bank
- Or prefer using **green energy over grid power**



WHERE?

Deployment in **High-Traffic Areas**:

- Schools, Train Stations, Airports, Parks, and Shopping Malls
- Outdoor Events & Exhibitions



WHO?

- **Individual Users (B2C)**: Students, professionals, tourists, and everyday commuters.
- **Implementing Entities (B2B/B2G)**: Corporations, educational institutions, event agencies, and city governments.



WHY?

- Integrated On-site Charging Hubs
- Tailored for the Vietnamese Market
- Pedestrian Traffic Tracking
- Eco-Centric Communication



HOW?

Step 1: User steps on the mat -> applies pressure to piezoelectric sensors.
Step 2: Mechanical energy is converted into electrical energy -> stored and stepped up.
Step 3: Stable power output is supplied via charging ports.
Step 4: OLED display shows real-time power metrics and charging status.
Step 5: Network-connected system collects and analyzes usage data.



Problem +
Solution

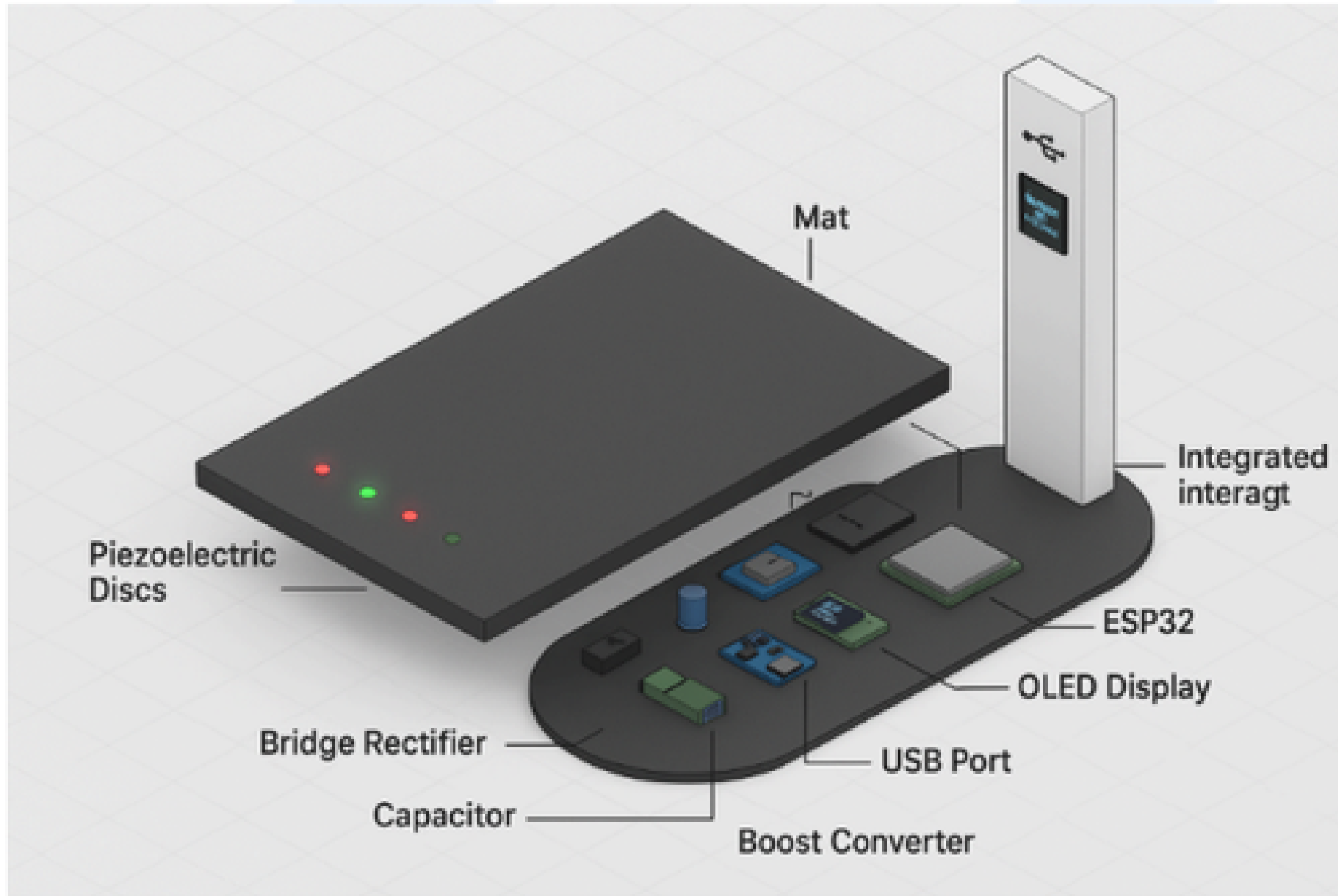
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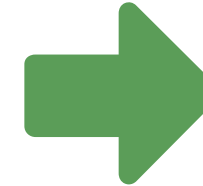
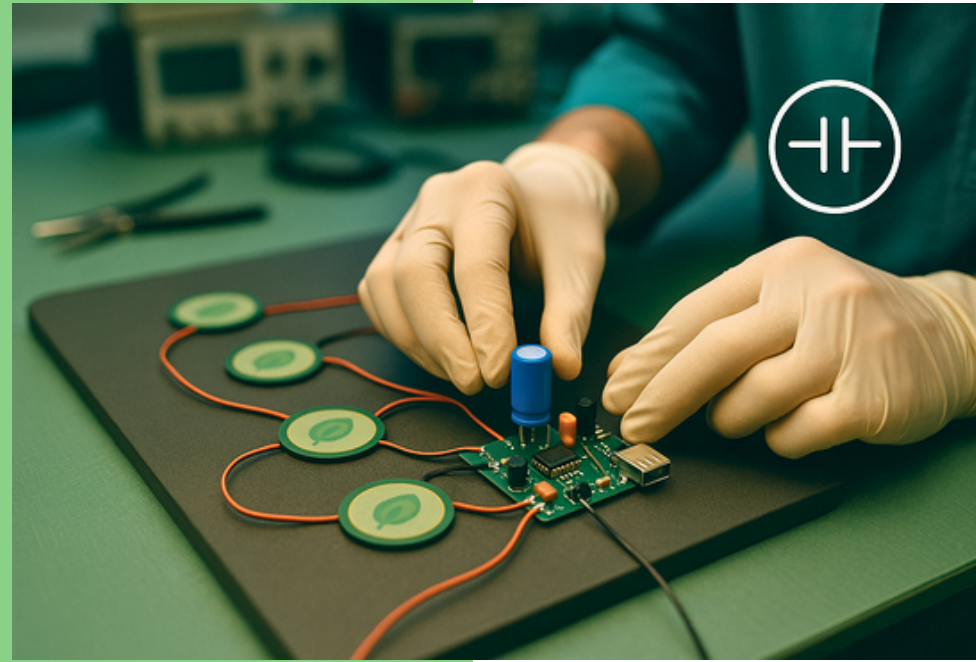
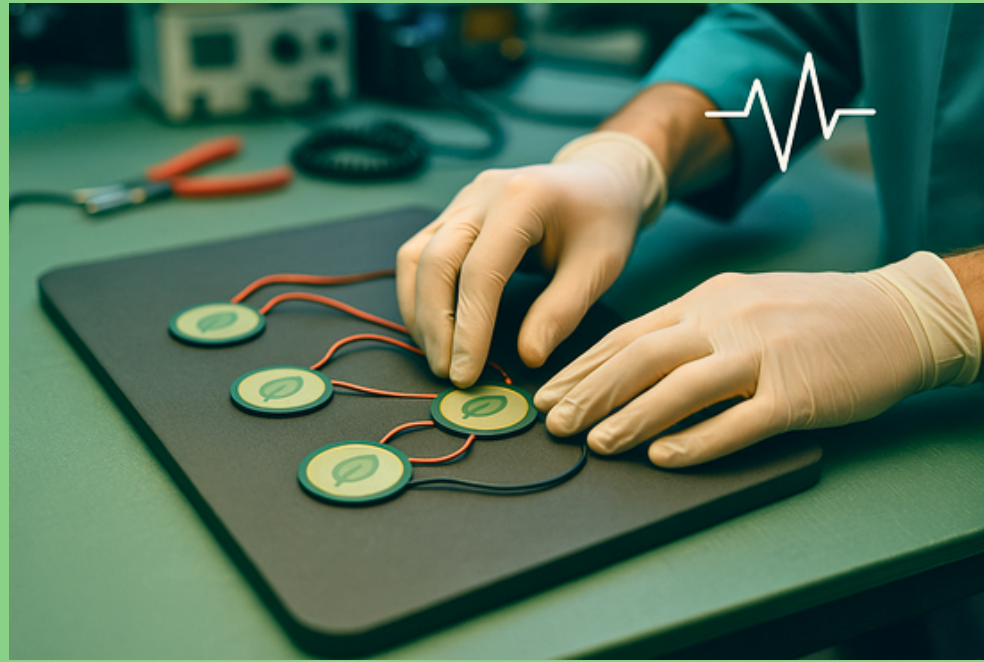
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GIAO DIỆN / KỸ THUẬT



Problem +
Solution

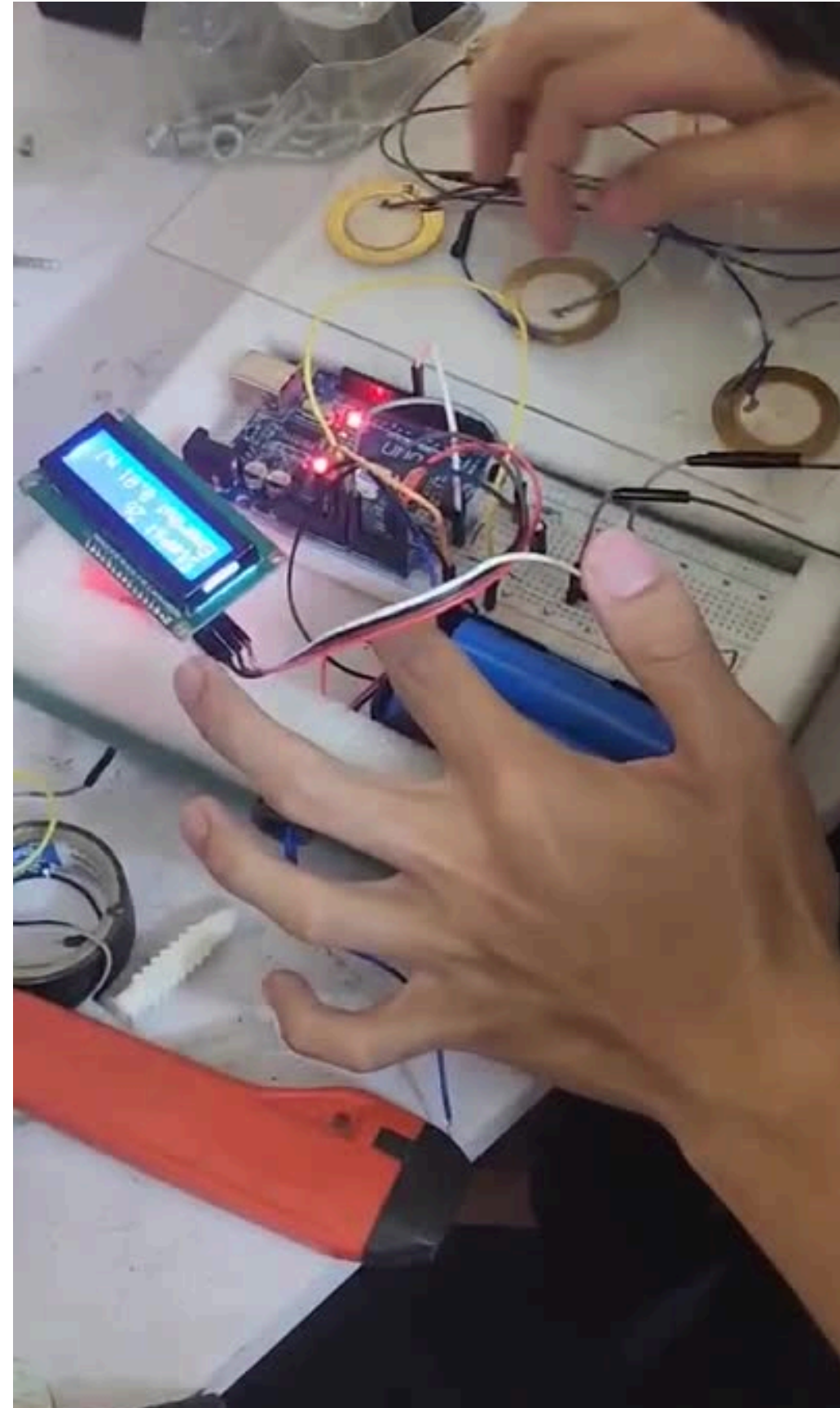
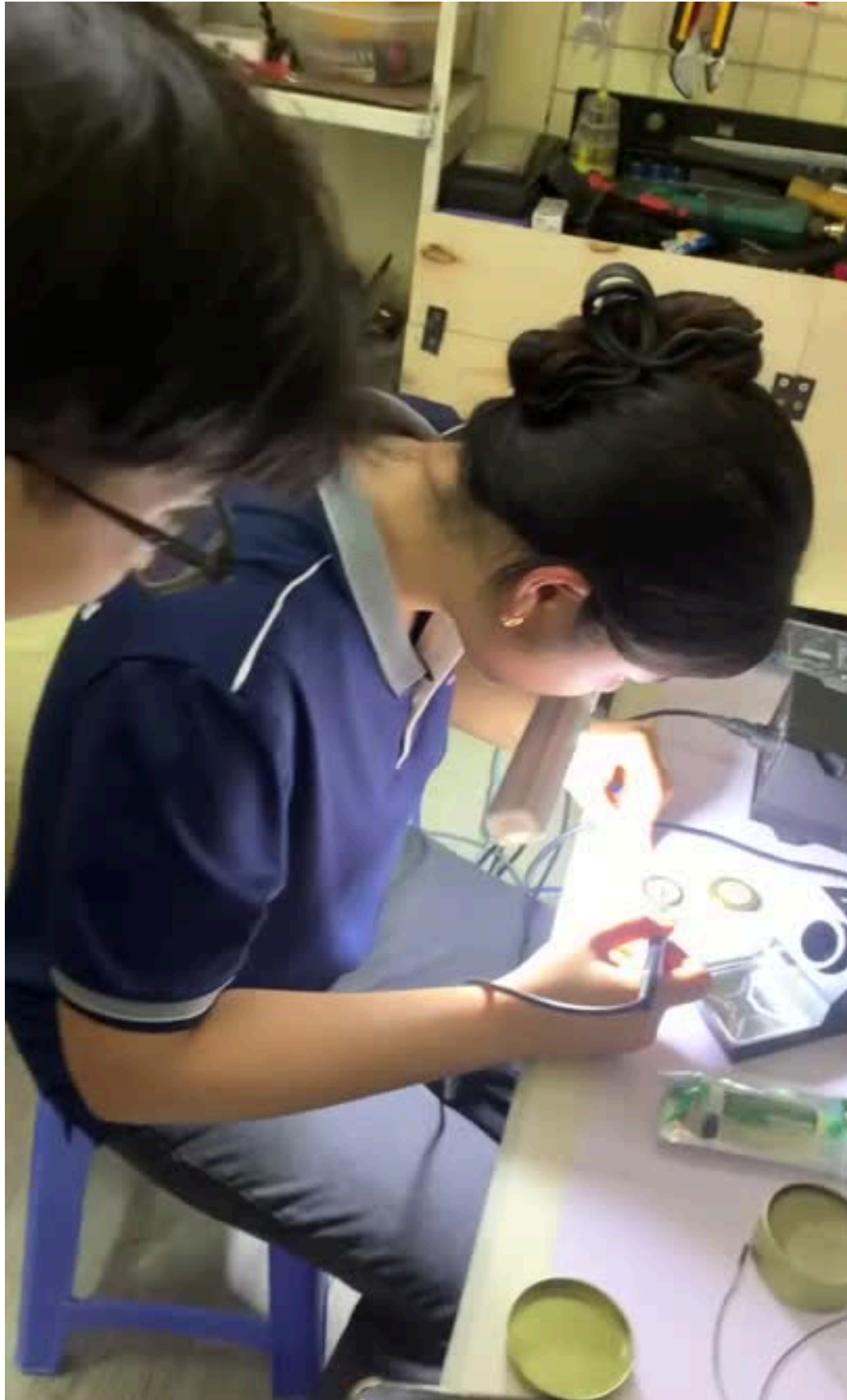
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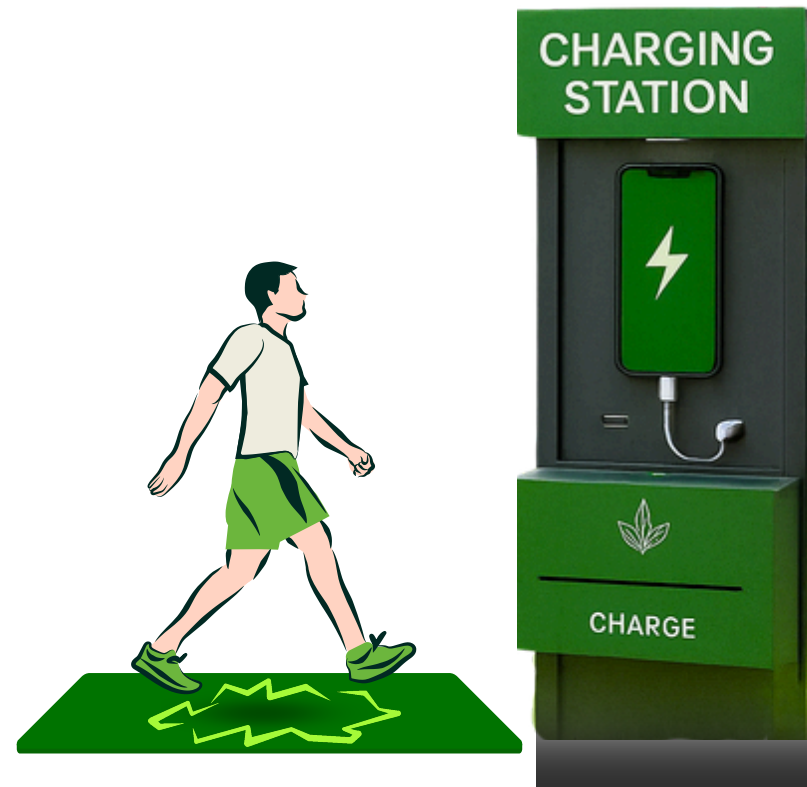
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0.0002 Wh



~1.7 mins of foot traffic



= +1% smartphone charge

TARGET CUSTOMERS & VALUE PROPOSITION

Airports

C: An engaging, green experience during tedious wait times.

B: Alleviates ESG compliance pressure



Educational Institutions

C: Combines physical activity with timely, on-the-go device charging.

B: Creates innovative "green spaces" to attract prospective students.

Events & Festivals

C: The harder fan, the brighter the lights.

B: Cuts temporary wiring costs and delivers a memorable experience.



Public Parks

C: Solves the lack of accessible power and utilities.

B: Transforms traditional parks into modern, safe "Energy Playgrounds."



MARKET SIZE



Total Addressable Market:

- Vast nationwide scale with over 2,700 potential locations.
- Capitalizing on the booming Green Transition and Smart City trends.

Serviceable Available Market:

- Focusing resources on 3 key cities: Hanoi, Ho Chi Minh City, and Da Nang.
- Approximately 1,399 target partners ready to invest within the first 3 years.

Serviceable Obtainable Market:

- Capturing 5% of SAM in the initial phase, securing 70 pioneer B2B clients.
- Maximizing revenue through a clustered deployment model (3-5 stations per location).

Problem +
Solution

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KEY PARTNERS

- Shopping malls, schools, airports
- Event organizers
- Government authorities & environmental organizations
- Manufacturing partners
- SMEs supplying energy storage equipment



KEY ACTIVITIES

- System development, installation, operation, and maintenance
- Communication & brand expansion
- Maintaining long-term partnerships



KEY RESOURCES

- R&D team
- Sensors, batteries, and controller equipment
- EcoCharge brand & interactive software



VALUE PROPOSITIONS

- Converting footsteps into electrical energy
- Integrated phone charging stations
- Modernizing public spaces
- Contributing to sustainable cities & green branding



CUSTOMER RELATIONSHIPS

- Long-term contracts (system sales/leasing)
- 24/7 technical support
- Communication & workshops



CHANNELS

- Direct B2B contracts
- Website & social media
- ESG communication campaigns



CUSTOMER SEGMENTS

- Corporate businesses & office buildings
- Schools & educational institutions
- Shopping malls, transit stations, and airports
- Event organizers



COST STRUCTURE

- R&D & upgrade costs
- Equipment & installation costs
- Personnel, marketing, and customer service
- Legal & initial communication costs



REVENUE STREAMS

- System sales or leasing
- Maintenance & software upgrade packages
- Foot traffic data services for malls, schools, and municipalities
- Sponsorships from CSR/ESG programs

Problem +
Solution

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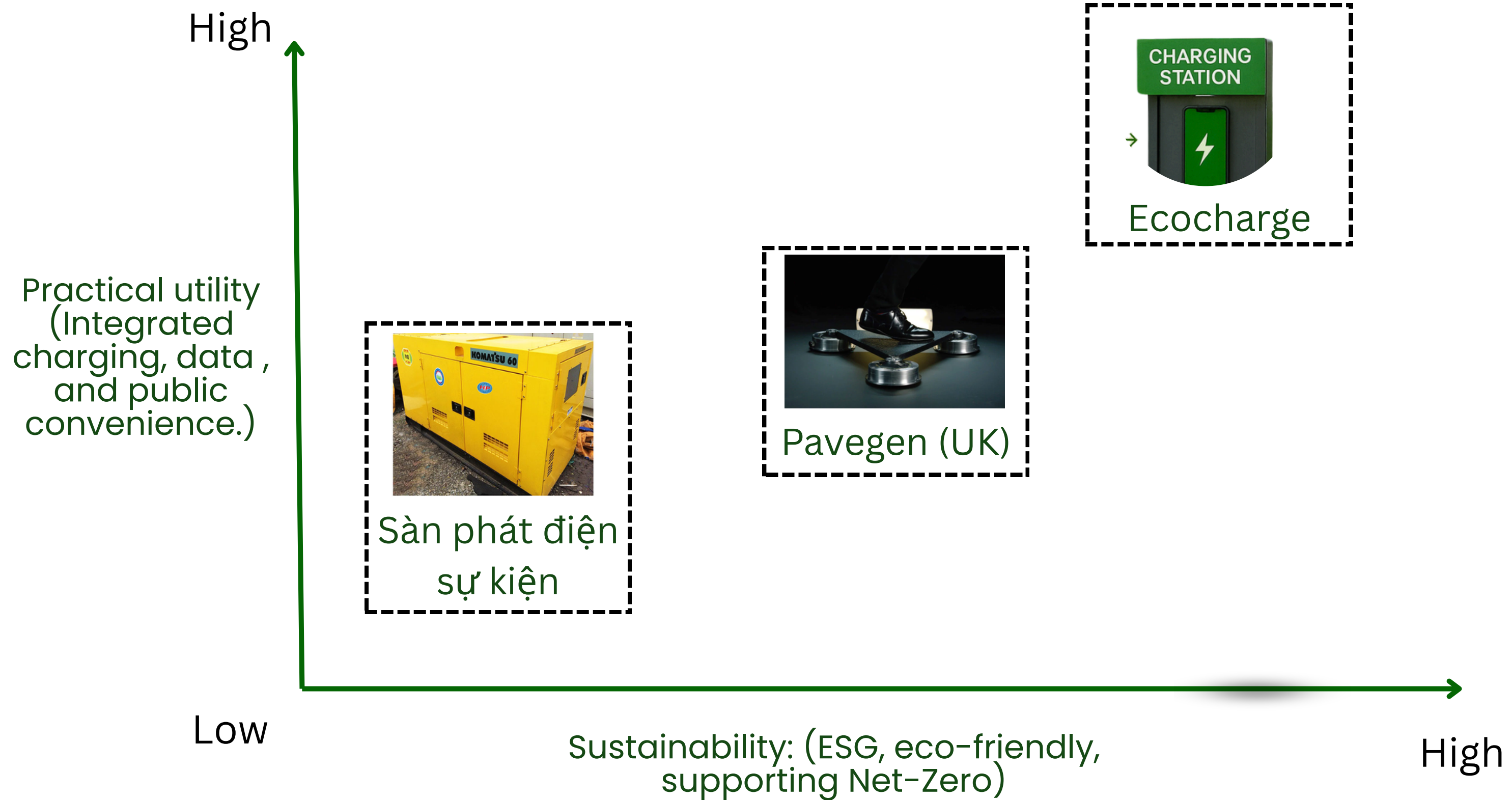
Business
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PRODUCT POSITIONING



MARKET ENTRY PLAN

Market Research & Preparation

- Assess demand & competitors
- Finalize product design
- Prepare materials & legal documents

Q1 2026

Market Testing

- Install 10–15 pilot stations
- Collect usage data
- Refine product & business model

Q2 2026

Market Rollout

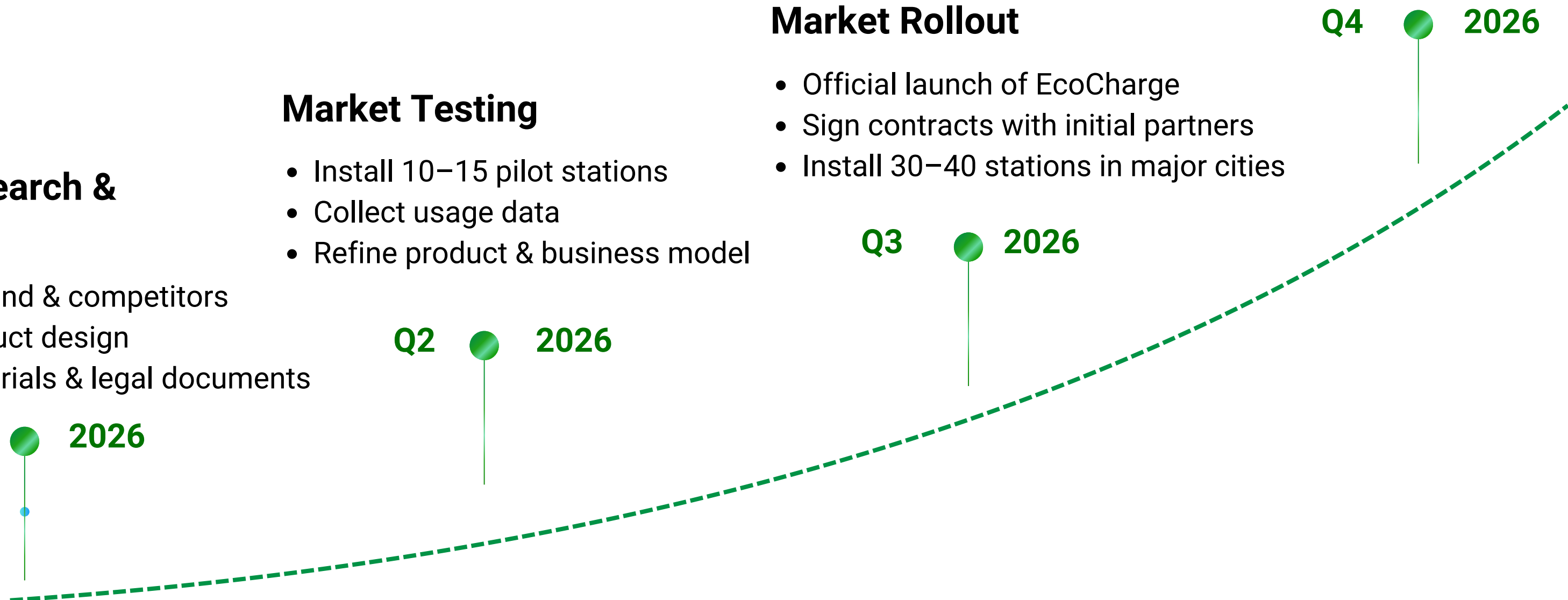
- Official launch of EcoCharge
- Sign contracts with initial partners
- Install 30–40 stations in major cities

Q3 2026

Expansion & Fundraising

- Summary report
- Prepare for Series A fundraising
- Plan for 2027 expansion (nationwide/regional)

Q4 2026



6-Month Detailed Communication Plan

Month	Core activitives	Action Items	Chanel	KPIs
1	Launch & Brand Awareness	"Step Forward" Launch Event (e.g., at HUST/Vincom) - Release TVC on YouTube & TikTok - Finalize brand identity	Events, Social Media, PR	200+ attendees 50,000 TVC views 1,000 new followers
2	Community Engagement	"1 Step = 1 Watt" photo campaign (station check-in) Distribute flyers at pilot stations - PR on major news outlets	Facebook, TikTok, PR online, offline	100,000 post reach 500 hashtag posts 3 major PR articles
3	Education & B2B Outreach	"GreenStep" school workshops "Calculate your generated power" online minigame - ESG-focused email marketing to businesses	Workshop offline, Email, Social media	3 workshops (100+ students) 1,000 minigame players 50 B2B email opens
4	Influencer & Community Collabs	KOL/Influencer experiences "Green Walk Day" community event - Release animated vision video	YouTube, TikTok, Outdoor Events	10 influencers 500,000 video views 1,000 event participants
5	Business Integration & CSR	"Sustainability in Motion" B2B seminar - Send sponsorship proposals (FMCG, Tech, Retail) - Launch EcoCharge station map on Google Maps	Seminars, Email, LinkedIn, Maps	5 B2B sponsors/pilots secured 1,000 map downloads/views
6	Evaluation & Re-strategizing	- Internal reports & user surveys at stations - Message repositioning (if needed) - Prepare expansion for HCMC, Da Nang	Survey, Dashboard, Team review	2,000 survey responses 70% positive feedback

ESG

SDG

E – ENVIRONMENT

- On-site clean energy generation
- Fossil fuel independence
- E-waste reduction
- Decrease the demand for portable power banks and external charging accessories

Measurable Impact:

- 5–7 mW/step → 2,000 people/day × 4 steps → 1.5–1.7 kWh/month per station.
- 0 kg CO₂ emissions (compared to natural gas at 2 kg CO₂/m³).
- Near-zero maintenance costs.

SDG 11 – Sustainable Cities and Communities:

Integrating renewable energy harvesting directly into existing urban infrastructure.



S – SOCIAL

- Free & safe utility
- Community education
- Raising environmental awareness
- Deploying in schools and malls → create interactive sustainability experiences for citizens

G – GOVERNANCE

- Transparent reporting
- Zero greenwashing
- Standardized inspection – maintenance processes
- Transparent leadership (CEO – COO – CTO – CMO – CFO)

SDG 13 – Climate Action:

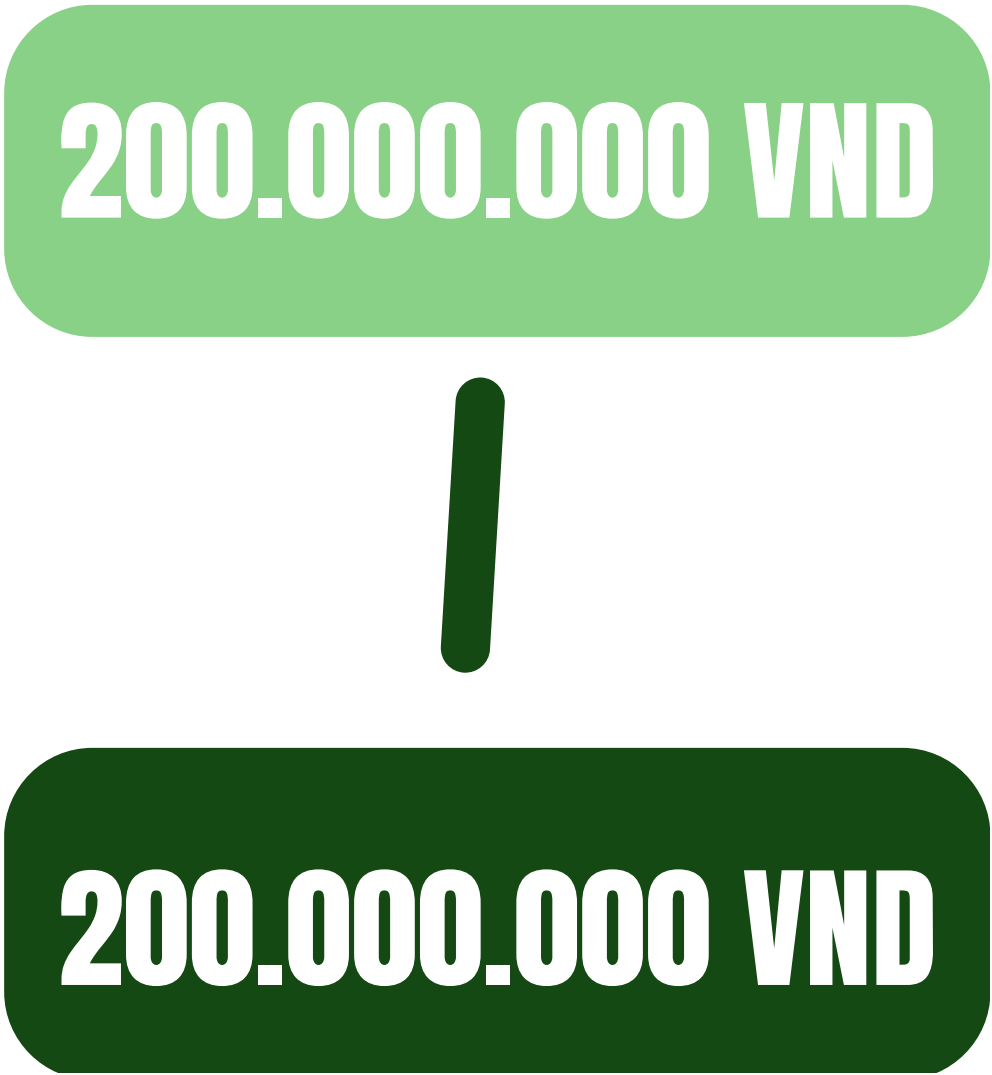
Actively reducing CO₂ footprint by incorporating green energy generation into daily public movement.



FUNDING OVERVIEW

Source of Funds	Amount (VND)	Ratio (%)
Founders' Contribution	100.000.000	20%
Investment Ask	400.000.000	80%
Total Capital	500.000.000	100%

2 FUNDING ROUNDS



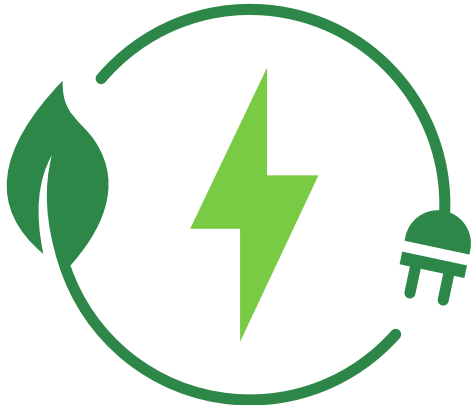
REVENUE

Revenue Streams	Charging Station Sales	Kinetic Mat Sales	Charging Station Leasing	Kinetic Mat Leasing
Unit Price (VND)	5,000,000	1,000,000	800,000	500,000
Revenue Mix (%)	10%	15%	30%	45%
Break-even Volume (Units)	94	141	282	422
Break-even Revenue (VND)	470,000,000	141,000,000	225,600,000	211,000,000
TOTAL	1,047,600,000			

Line Item	YEAR 1	YEAR 2	YEAR 3
(+) Revenue	50.000.000	1.200.000.000	3.000.000.000
(-) Cost of Goods Sold (COGS) Production & Installation Costs	20.000.000	540.000.000	1.350.000.000
(-) Operating Expenses (OPEX) R&D, Marketing, HR, Legal	430.000.000	195.000.000	225.000.000
(=) Earnings Before Tax (EBT)	(400.000.000)	465.000.000	1.425.000.000
(-) Corporate Income Tax (20%)	0	93.000.000	285.000.000
(=) Net Profit	(400.000.000)	372.000.000	1.140.000.000

REVENUE

Year 1: Planned strategic loss to focus on R&D and pilot testing.
Year 2: Achieving break-even point and initiating profitability through commercialization.
Year 3: High profitability driven by market scaling and economies of scale.



THANK YOU

*Every Step Powers the Future.
EcoCharge - Charge Green, Live Smart.*





